Integration Step

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October 13, 2026

Hypothesis :

$$(\int_{B_R} \mathrm{d}x\mathbb{E}^{\frac{2}{q}} \left|(\delta\Pi^- - \mathrm{d}\Gamma_x^*\Pi_x^-)_{\beta r}(x)\right|^q)^{\frac{1}{2}} \lesssim r^{\alpha-2+\frac{D}{2}}(r+R)^{|\beta|-|\alpha|}, \tag{1}$$

$$(\int_{B_R} \mathrm{d}x\mathbb{E}^{\frac{2}{q}} \left|(\mathrm{d}\Gamma_x^*)_\beta^\gamma\right|^q)^{\frac{1}{2}} \lesssim R^{\frac{D}{2}+|\beta|-|\gamma|} \text{ for populated } \gamma \neq \mathrm{pp}, \tag{2}$$

$$\partial^{\mathbf{n}}((\delta\Pi - \mathrm{d}\Gamma_x^*\Pi_x)_\beta)(x) = 0,\ \forall x \in \mathbb{R}^d, \forall\, |\mathbf{n}| < 2+s \tag{3}$$

Goal :

$$\left(\int_{B_R} \mathrm{d}x\mathbb{E}^{\frac{2}{q}} |(\delta\Pi - \mathrm{d}\Gamma_x^*\Pi_x)_{\beta r}(x)|^q\right)^{\frac{1}{2}} \lesssim r^{\alpha+\frac{D}{2}}(r+R)^{|\beta|-|\alpha|} \tag{4}$$

Lemma 1 (Liouville)**.** *Let g be a smooth function, $m \in \mathbb{N}$ an integer, and $x \in \mathbb{R}^n$ a base point. Then there exists at most on couple (u,p), with u a smooth function and p a polynomial, such that :*

$$\begin{cases} L(u) = g - p \\ \deg(p) \leq m-2 \\ \forall\, |\mathbf{n}| \leq m, \partial^{\mathbf{n}}u(x) = 0. \limsup_{|x-y|\to\infty} \frac{u(y)}{|x-y|^\kappa} < \infty, \text{ for } \kappa \in \mathbb{R},\ \kappa < m. \end{cases}$$

Proof. Let $|\mathbf{n}| \leq m-2$.

$$0 = \partial^{\mathbf{n}}(L(u))(x) = \partial^{\mathbf{n}}g(x) - \partial^{\mathbf{n}}p(x),$$

□

so that p is determined by g. Then, let u and $\tilde{u}$ be two candidate and denote $w = u - \tilde{u}$. We have $L(w) = 0$, and $\limsup_{|x-y|\to\infty} \frac{u(y)}{|x-y|^\kappa} < \infty$. By Liouville's theorem, w is a polynomial with degree at most κ. Then $\forall\, |\mathbf{n}| \leq m, \partial^{\mathbf{n}}w(x) = 0$ shows that $w = 0$.

We have $L(\delta\Pi_\beta - (\mathrm{d}\Gamma_x^*\Pi_x)_\beta) = \delta\Pi_\beta^- - (\mathrm{d}\Gamma_x^*\Pi_x^-)_\beta \bmod \mathcal{P}_{<|\beta|+\frac{D}{2}-2}$. But it is not sufficient to caracterize $\delta\Pi_\beta - (\mathrm{d}\Gamma_x^*\Pi_x)_\beta$ in terms of $\delta\Pi_\beta^- - (\mathrm{d}\Gamma_x^*\Pi_x^-)_\beta$, because the growth at infinity of $\delta\Pi_\beta - (\mathrm{d}\Gamma_x^*\Pi_x)_\beta$ is too big. Indeed, $(\mathrm{d}\Gamma_x^*\Pi_x)_\beta = \sum_{|\gamma|<|\beta|+\frac{D}{2}} (\mathrm{d}\Gamma_x^*)_\beta^\gamma \Pi_{x\gamma}$. Now $\Pi_{x\gamma}(y)$ is expected to grow like $|x-y|^\gamma$. So the sum include terms that grow almost like $2+\frac{D}{2}$. In contrast, the vanishing order of $\delta\Pi_\beta - (\mathrm{d}\Gamma_x^*\Pi_x)_\beta$ at x would be $\alpha+\frac{D}{2}(=2+s)$, which is lower by more than 2 (we are in the regime $\alpha < 0$). This is the reason why we truncate the sum at order $2+s$. Furthermore, because $L\Pi_{x\beta} = \Pi_{x\beta}^- \bmod \mathcal{P}_{<|\beta|-2}$ and $L\delta\Pi_\beta = \delta\Pi_\beta^- \bmod \mathcal{P}_{<|\beta|-2}$, we restrict ourselves to $|\beta| < \lceil 2+s \rceil$. As a result, we have the following

$$L(\delta\Pi_\beta - \sum_{|\gamma|<2+s} (\mathrm{d}\Gamma_x^*)_\beta^\gamma \Pi_{x\gamma}) = \delta\Pi_\beta^- - \sum_{|\gamma|<2+s} (\mathrm{d}\Gamma_x^*)_\beta^\gamma \Pi_{x\gamma}^- \bmod \mathcal{P}_{<s} \tag{5}$$

and the hypothesis of the lemma are verified, so $U_x \coloneqq \delta\Pi_\beta - \sum_{|\gamma|<2+s}(\mathrm{d}\Gamma^*_x)^\gamma_\beta \Pi_{x\gamma}$ is completely determined by $F_x \coloneqq \delta\Pi^-_\beta - \sum_{|\gamma|<2+s}(\mathrm{d}\Gamma^*_x)^\gamma_\beta \Pi^-_{x\gamma}$.

Remark 2. *The threshold $2+s$ is too strong for $(\mathrm{d}\Gamma^*_x\Pi^-_x)_\beta$, but we will need to deal with the remainder $\sum_{|\gamma|\geq 2+s}(\mathrm{d}\Gamma^*_x)^\gamma_\beta\Pi^-_{x\gamma}$ and see that this threshold makes sense.*

We claim that (2) implies

$$\left(\int_{B_R} \mathrm{d}x\mathbb{E}^{\frac{2}{q}} \left|(\sum_{|\gamma|\geq 2+s}(\mathrm{d}\Gamma^*_x)^\gamma_\beta\Pi^-_{x\gamma})_r(x)\right|^q\right)^{\frac{1}{2}} \lesssim r^{\alpha+\frac{D}{2}-2}(r+R)^{|\beta|-|\alpha|}, \tag{6}$$

$$\left(\int_{B_R} \mathrm{d}x\mathbb{E}^{\frac{2}{q}} \left|(\sum_{|\gamma|\geq 2+s}(\mathrm{d}\Gamma^*_x)^\gamma_\beta\Pi_{x\gamma})_r(x)\right|^q\right)^{\frac{1}{2}} \lesssim r^{\alpha+\frac{D}{2}}(r+R)^{|\beta|-|\alpha|}. \tag{7}$$

So that by triangle inequality, we can add to our set of hypothesis

$$\left(\int_{B_R} \mathrm{d}x\mathbb{E}^{\frac{2}{q}} |(F_x)_r(x)|^q\right)^{\frac{1}{2}} \lesssim r^{\alpha+\frac{D}{2}-2}(r+R)^{|\beta|-|\alpha|},$$

and it suffices to show

$$\left(\int_{B_R} \mathrm{d}x\mathbb{E}^{\frac{2}{q}} |(U_x)_r(x)|^q\right)^{\frac{1}{2}} \lesssim r^{\alpha+\frac{D}{2}}(r+R)^{|\beta|-|\alpha|}.$$

The strategy now is to take advantage of the unicity of solution to $Lu = F_x \bmod \mathcal{P}_{<s}$, to find another expression of U_x, that verifies (4). The germ (F_x) is $(2\alpha, \alpha+\frac{D}{2}-2)$-coherent and $\alpha+\frac{D}{2}-2$ - homogeneous. Let K be the heat kernel, which is assumed to be a 2-regularising kernel (we would need to truncate it, for it to be compactly supported, but it is easy to deal with the smooth remainder). Define $\tilde{U}_x \coloneqq \mathcal{K}(F)_x = K*(F_x) - T^{2+s}_x(K*(F_x))$. By the Schauder estimate theorem for coherent germs (Theorem 3.17 of [1]), $\tilde{U}_x$ is a well-defined $(2\alpha+2, \alpha+\frac{D}{2})$-coherent and $\alpha+\frac{D}{2}$-homogeneous germ. Plus its homogeneity norm is dominated by the sum of the homogeneity and coherence norm of (F_x). This means that

$$\left(\int_{B_R} \mathrm{d}x\mathbb{E}^{\frac{2}{q}} \left|(\tilde{U}_x)_r(x)\right|^q\right)^{\frac{1}{2}} \lesssim r^{\alpha+\frac{D}{2}}((C+R)^{|\beta|-\alpha} + (C+R)^{|\beta|-3\alpha-2}) \tag{8}$$

$$\lesssim r^{\alpha+\frac{D}{2}}(C+R)^{|\beta|-\alpha} (\text{because } 3\alpha+2>\alpha) \tag{9}$$

which is exactly the estimate we want to prove for (U_x).

Let's show that $\tilde{U}_x = U_x$. It suffices to check that both $\tilde{U}_x$ and U_x verifies Liouville's lemma's hypothesis. First, $L\tilde{U}_x = L(K*(F_x)) - L(T^{2+s}_x(K*(F_x))) = F_x +$ polynomial of degree $<$ s. Then,

- The jet condition is verified by construction of $\tilde{U}_x$.
- With (7), we claimed that the remainder $\sum_{|\gamma|\geq 2+s}(\mathrm{d}\Gamma^*_x)^\gamma_\beta\Pi_{x\gamma}$ has vanishing order $\alpha+\frac{D}{2} = 2+s$ at x, and together with (3), it shows that the jet condition is also verified for U_x.
- We already dealt with the growth condition of U_x, it was the reason we truncated.
- Growth condition for $\tilde{U}$?

which concludes.

It remains to prove that (2) implies (7) and (6). This is done purely by induction.

The sums $\sum_{|\gamma|\geq 2+s}(\mathrm{d}\Gamma_x^*)_\beta^\gamma \Pi_{x\gamma}$ and $\sum_{|\gamma|\geq 2+s}(\mathrm{d}\Gamma_x^*)_\beta^\gamma \Pi^-_{x\gamma}$ is restricted to populated γ that are not purely polynomial : else $\Pi^-_{x\gamma}=0$, and $(\mathrm{d}\Gamma_x^*)_\beta^\gamma = (\mathrm{d}\Gamma_x^*)_\beta^{\delta_{\mathbf{n}}} = 0$ because $|\mathbf{n}|\geq 2+s$. Furthermore, $|\gamma|_\prec < |\beta|_\prec$ so the induction hypothesis on $\Pi^-_{x\gamma}$ and $\Pi_{x\gamma}$ are available.

$$\begin{aligned}
\left(\int_{B_R} \mathrm{d}x \mathbb{E}^{\frac{2}{q}} \left| (\sum_{|\gamma|\geq 2+s} (\mathrm{d}\Gamma_x^*)_\beta^\gamma \Pi^-_{x\gamma})_r(x)\right|^q\right)^{\frac{1}{2}} &\leq \sum_\gamma \left(\int_{B_R} \mathrm{d}x \mathbb{E}^{\frac{2}{q}} \left|((\mathrm{d}\Gamma_x^*)_\beta^\gamma \Pi^-_{x\gamma})_r(x)\right|^q\right)^{\frac{1}{2}} \text{(triangle inequality)} \\
&\leq \sum_\gamma \left(\int_{B_R} \mathrm{d}x \mathbb{E}^{\frac{2}{2q}} \left|(\mathrm{d}\Gamma_x^*)_\beta^\gamma\right|^{2q}\right)^{\frac{1}{2}} \left(\int_{B_R} \mathrm{d}x \mathbb{E}^{\frac{2}{2q}} \left|(\Pi^-_{x\gamma})_r(x)\right|^q\right)^{\frac{1}{2}} \text{(Höl} \\
&\leq \sum_{2+s\leq|\gamma|\leq|\beta|+\frac{D}{2}} R^{\frac{D}{2}+|\beta|-|\gamma|} r^{\gamma-2} \\
&\lesssim r^{\alpha-2+\frac{D}{2}} (r+R)^{|\beta|-\alpha}.
\end{aligned}$$

And we proceed the same way for (7).

References

[1] Lucas Broux, Francesco Caravenna, and Lorenzo Zambotti. "Hairer's Multilevel Schauder Estimates without Regularity Structures". In: *Transactions of the American Mathematical Society* (Aug. 9, 2024). ISSN: 0002-9947, 1088-6850. DOI: 10.1090/tran/9245. arXiv: 2301.07517 [math.AP]. URL: http://arxiv.org/abs/2301.07517 (visited on 06/16/2026).